

### 3.3.1 Number of research papers published per teacher in the Journals as Notified on UGC CARE list during the Academic Year 2022

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| 2    | Critical Success Factors of Total Quality Management Implementation In Indian Iron And Steel Industries (IISI)                                               | Mr. Haldhar Ram Chandrakar                                           |
| 3    | Functionally Graded Carbon Nanotubes Reinforced composite Structures: An Extensive Review                                                                    | Dr.Sourabh Kumar Soni                                                |
| 4    | Production of Bamboo Fibre Reinforced Epoxy Hybrid Composite with Nano sio2 Particles                                                                        | Dr. AbhijeetGanguly                                                  |
| 5    | Improvement of Overall Equipment Effectiveness Using Total Productive Maintenance and Failure Mode Effect Analysis in Blast Furnace of Iron & Steel Industry | Dr. Rajesh Kumar1,Dr. Abhijeet Ganguly2,Mr. Revendra Kumar Deshmukh3 |
| 6    | Review of Total Productive Maintenance Using Failure Mode Effect Analysis across Various Industries                                                          | Dr. Rajesh Kumar1,Dr. Abhijeet Ganguly2,Mr. Revendra Kumar Deshmukh3 |
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| 12   | A Simulation Of A Statcom-Control For Grid Connected Wind Energy System For Improving Reliability Of Power                                                   | Dr.Padmini Sharma                                                    |
| 13   | Fake Review Prediction of Online Product: A Survey                                                                                                           | Mr. Prabhas Kumar Gupta,                                             |



**Recent Trends in Product Design and Intelligent Manufacturing Systems** pp  
749–758

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paper

## Analysis of Productivity and Surface Characteristics of A356-TiB<sub>2</sub> Nanocomposites in EDM

[Shallesh Dewangan](#) , [Santosh Kumar Sharma](#), [Amit Kumar  
Vishvakarma](#) & [Chitrakant Tiger](#)

Conference paper | [First Online: 06 October 2022](#)

407 Accesses

Part of the [Lecture Notes in Mechanical Engineering](#) book  
series (LNME)

### Abstract

The manufacturing industry across the globe is accelerating at a rapid pace, and India is emerging as a preferred hub for manufacturing of automobile and aerospace components. Aero-components involve high-performance metal alloys and composites. Due





Review

# Functionally graded carbon nanotubes reinforced composite structures: An extensive review

Sourabh Kumar Soni<sup>a</sup>, Benedict Thomas<sup>a</sup> , Ashirbad Swain<sup>a</sup>, Tarapada Roy<sup>b</sup> 

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## Abstract

In the past few years, a significant research effort has been imposed by the researchers on the in-depth analysis of functionally graded carbon nanotubes reinforced composite (FG-CNTRC) structures due to their superior mechanical, thermal, optical, thermo-mechanical and vibration characteristics over conventional composites. This study presents a comprehensive review of the reported studies on the mechanical, vibration, thermal, thermo-mechanical and low-velocity impact responses of FG-CNTRC structures subjected to numerous loads, end supporting conditions and different environments. An attempt has been made here, to include both experimental and theoretical studies on the investigation of the FG-CNTRC structures, along with the challenges and critical areas regarding future research. This study includes: (i) a brief introduction to composites, functionally graded materials (FGMs), and nanocomposites; (ii) the antiquity of carbon nanotubes (CNTs); (iii) an introduction to FG-CNTRC; (iv) a comprehensive review of mechanical, vibration, thermal, thermo-mechanical, viscoelastic properties and low-velocity impact analysis of FG-CNTRC structures (plate/beam/shell/panels); and (v) a discussion on the issues/challenges and possible scope for the future research on the FG-CNTRC.

## Introduction

Intensifying demands of high-strength new materials with enhanced service life, reduced weight and economic features pledge the development and application of new materials. It is a well-known fact that in the design or fabrication of any component or structure, materials properties play a crucial role. Usually, the

## Production of Bamboo Fibre Reinforced Epoxy Hybrid Composite with Nano SiO<sub>2</sub> Particles

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This paper focus on the process of fiber reinforced composites a significant portion of the paper emphasis the use of different machine, basic concept introduced in this paper of molding method and pouring of epoxy, hardener and nano silica mixture with hand lay-up technique. This study aims to develop fibre-reinforced epoxy hybrid composites by using hand lay-up method. In this work different material used are Epoxy LY556, Hardner HY951, SiO<sub>2</sub> nano particle (Particle size: 60-70nm) and 90 degree bidirectional woven bamboo fibre. Sample designated are as A10 ( One layer Bubble fibre, 0% SiO<sub>2</sub>), A12 (2%SiO<sub>2</sub>), A14 (4%SiO<sub>2</sub>), B20 (Two-layer bamboo fibre, 0% SiO<sub>2</sub>), B22(2%SiO<sub>2</sub>), B24(4%SiO<sub>2</sub>), C30 (three layer bamboo fibre, 0% SiO<sub>2</sub>), C32(2%SiO<sub>2</sub>) and C34(4%SiO<sub>2</sub>). Samples were prepared by handlayup method. The effect of various weight percent of SiO<sub>2</sub> on mechanical property were investigated. The maximum value of tensile strength was found for sample C32, higher value of flexural strength was found for sample C32, higher value of izod impact test was for sample C32 and maximum value of barcol hardness was for sample B22.

**Keywords:** Natural-Fibres, Bamboo fibre reinforced nano composites, Hand lay -up technique, Epoxy matrix, Hardener

## 1 Introduction

Composite material is formed by mixing two or more material, the main component of which may be epoxy, hardener and the properties of the material to be made the details of the material to be made by hand lay-up technique the mechanical properties of the substance have also been widely investigated. In today's era composite materials are being used in abundance such as it is also used in home use, automotive application air craft, industries and sporting goods at present, the composite materials are being developed using natural fibre because natural fibre composites are relative eco-friendly. Some research on the fabrication method of composite has been done. Hand lay-up is that the simplest and oldest open molding technique for development of composites. Jindal<sup>1</sup> *et al.* explained in his work emphasized on the work with natural fiber at present engineers and scientists working on this area mainly utilizing natural fiber, developing environment friendly materials. Gunasekaran *et al.*<sup>2</sup> studied and presented useful analysis, which is important for quality development of bamboo based organic compound for different applications. Analysis of changes and traits from

mixing bamboo charcoal particles with other fibre sources is essential. Kim *et al.*<sup>3</sup> presented the method in his paper Hand Lay-up, which is an easiest process with low investment, comparably favourably with resin transfer molding processes and Vacuum Infusion which require high tooling cost. However, the technology, which uses vacuum pressure to force resin into a composite laminate, better structure of GFRP with higher properties which is said to be used in ship technology. Sharma and Wetzel<sup>4</sup> discussed, the techniques of manufacturing of GFRP structure include the fabrication method hand lay-up, filament winding, vacuum infusion and resin transfer molding. Kumar *et al.*<sup>5</sup> performed experiments which includes analysis of tensile properties of natural bamboo fibres. It consists calculation of tensile strength of laminate bamboo fibre with sodium hydroxide. The development of various bamboo fibre hybrid composites samples was made by hand layup method. Study shows that the tensile strength of bamboo-epoxy composite increases to the certain level of fibre loading and then starts decreasing on further fibre loading. On basis of obtain value tensile strength of bamboo fibre is high<sup>6,7</sup>. Various researchers have used the herbal fibers containing polyolefin's, polystyrene, polyester and epoxy resins. Vikram *et al.*<sup>8</sup> reported that natural bamboo fiber strengthened epoxy-primarily based

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# Improvement of Overall Equipment Effectiveness Using Total Productive Maintenance and Failure Mode Effect Analysis in Blast Furnace of Iron & Steel Industry

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**Abstract**— This paper aims for conducting Total Preventive Management significantly using Failure Mode Effect Analysis in Blast Furnace to improve productivity quality and reliability of Iron and Steel Industries. Many organisation experience a halt in operations as a result of damaged production machines; as a result, customer demand is not met, and productivity suffers. This study is conducted at Jaiswal NICO private limited, Raipur. Blast furnace where general equipments are very delicate and need to be replaced frequently. It results in slow down of operation. Calculation of overall equipment effectiveness is affected due to low efficiency and lack of continuous analysis. TPM-FMEA technique was much required for blast furnace to eradicate the general failure. In this paper first the general problem where identified for various equipment in blast furnace, the TPM is done and OEE is calculated. Again FMEA is applied to the same process and OEE is calculated again. Results are compared for values of OEE before FMEA (Financial Year 2018-19) and after FMEA (Financial Year 2019-20). Results are being validated using SPSS Software. Value of Overall Equipment Effectiveness is more in financial year 2019-20. After FMEA four components were identified which have high failure rate. Six big losses are also calculated to identify the time losses more minutely. TPM-FMEA is more comprehensive than alone TPM or alone FMEA. It gives more detail on failure root cause and potential failure also. During maintenance TPM-FMEA can be applied quarterly or six monthly so as to reduce the utilization losses and speed losses from time to time.

**Keywords:** FMEA, TPM, Quality, Reliability, Productivity, OEE, RPN, SPSS

## I. INTRODUCTION

Total Productive Maintenance (TPM) is a unique Japanese philosophy, which has been developed based on the productive maintenance concepts and methodologies. This concept was first introduced by M/s Nippon Denso Co. Ltd. of Japan, a supplier of M/s Toyota Motor Company, Japan in the year 1971. Total productive maintenance is an innovative approach to maintenance that optimizes equipment effectiveness, eliminates breakdowns and promotes autonomous maintenance by operators through day-to-day activities involving total workforce.

Nakajima (1989), a major contributor of TPM, has defined TPM as an innovative approach to maintenance that optimizes equipment effectiveness, eliminates breakdowns, and promotes autonomous maintenance by operators through day-to-day activities involving the total workforce. In this stage eight activities are carried which are called eight pillars in the development of TPM activity. Of these four activities

are for establishing the system for production efficiency, one for initial control system of new products and equipment, one for improving the efficiency of administration and are for control of safety, sanitation as working environment.

## A. Eight Pillars of TPM

- 1) 5S: TPM starts with 5S. Problems cannot be clearly seen when the work place is unorganized. Cleaning and organizing the workplace helps the team to uncover problems. Making problems visible is the first step of improvement. The meaning of 5S is SEIRI- Sort out, SEITON - Organize, SEISO - Shine the workplace, SEIKETSU - Standardization, SHITSUKE - Self-discipline.
- 2) Autonomous Maintenance : This pillar is geared towards developing operators to be able to take care of small maintenance tasks, thus freeing up the skilled maintenance people to spend time on more value added activity and technical repairs. The operators are responsible for upkeep of their equipment to prevent it from deteriorating.
- 3) Continuous Improvement (KAIZEN): Kai" means change, and "Zen" means good (for the better). Basically kaizen is for small improvements, but carried out on a continual basis and involve all people in the organization. Kaizen is opposite to big spectacular innovations. Kaizen requires no or little investment. The principle behind is that "a very large number of small improvements are more effective in an organizational environment than a few improvements of large value. This pillar is aimed at reducing losses in the workplace that affect our efficiencies. By using a detailed and thorough procedure we eliminate losses in a systematic method using various Kaizen tools. These activities are not limited to production areas and can be implemented in administrative areas as well.
- 4) Planned Maintenance: It is aimed to have trouble free machines and equipments producing defect free products for total customer satisfaction. This breaks maintenance down into 4 "families" or groups such as Preventive Maintenance, Breakdown Maintenance, Corrective Maintenance, Maintenance Prevention
- 5) Quality Maintenance: It is aimed towards customer delight through highest quality through defect free manufacturing. Focus is on eliminating non-conformances in a systematic manner, much like Focused Improvement. We gain understanding of what parts of the equipment affect product quality and begin to eliminate current quality concerns, then move to potential quality concerns.



## REVIEW OF INTEGRATED APPROACH OF LEAN, SIX SIGMA AND THEORY OF CONSTRAINTS

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### Abstract

In any organization Continuous Improvement methods are developed and deployed with some standards to a sustained period to gain profitability, increase market share, and improve brand value and to survive in the market. In this research, an attempt is made for demonstrating the integrated concept of Lean & Six Sigma and Lean & Six Sigma and Theory of Constraints to get the maximum throughput which is the primary goal of any Industry. This paper review the practical implication of Lean management, Six sigma and Theory of Constraints in industry. The globalization and competitive business environment creates a hunt for new sources of sustainable developments. It is investigated to show the research carried out for efficiency improvement on the performance achievements of the team goals. There can be various method to calculate the ranking based on various performance levels.

**Keywords:** FMEA, TPM, Quality, Reliability, Productivity, TOC, Lean Management, Six Sigma.

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### 1. INTRODUCTION

The current cut-throat business scenario forces the industries to innovate and improve process to accelerate the product development. The process improvement methods implemented in product development helps in quality, velocity and sustainability in long run. The coordination between the industries and research are the need for the current industries to improve in their process efficiency. Detailed literature survey is carried out on the process improvement methods implemented in industries across different domains. Continuous improvements methods are concerned with the management organizational activities for improving the sequence of process. There is a steep increase in the implementation of the combined approach of methods for the process improvement by the industries. Researchers conduct lot of literature in the field of integrated approach for the continuous improvements methods to carry out their research and conclude the benefits (Dave Nave, 2002). Many researchers carried out their work in the field of process improvement and industries to spend millions of dollars in the process improvements.

## AN INTEGRATED APPROACH FOR RANKING OF EMPLOYEES APPRAISAL RATING: A CASE STUDY

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### Abstract

The search for new avenues of sustainable developments is driven by globalization and the competitive economic environment. Any organization that wants to be profitable, gain market share, enhance brand value, and remain competitive must create and implement Continuous Improvement techniques that adhere to certain criteria over an extended period of time. In this research, an effort is aimed at demonstrating how Lean, Six Sigma and Theory of Constraints may work together to achieve maximum throughput, which is every industry's top priority. Further research is done to demonstrate the efficiency gain in performance successes of the team goals. Finally, a comparison of the two integrated approaches' levels of efficiency improvement is done.

An integrated methodology employing lean and six sigma is recommended at the initial stage of the task. For this reason, the process improvement throughout product development of the heavy engine team objective achievements is taken into consideration as a case study example. In order to demonstrate the advancement of process improvement, performance data for the individual goals of a team are gathered and a statistical approach is put into use. The second stage of the project involves the proposal of an integrated methodology that combines lean, six sigma, and theory of constraints. To forecast the teams and the individual team members' performance metrics, an empirical model is developed. Through an iterative process, the weights for each team member's individual goals are allocated and updated. Each employee's and team's grade ranks are calculated and reported using an iterative process. The team and the employees gain insight over how to improve their performance. Furthermore, this method to constant improvement inspires confidence in the top management.

With the use of the Lean & Six Sigma methodology, it is inferred from the research data that the objective rate has improved. Moreover, it has been determined that the effectiveness of the objective achievements using the theory of constraints, lean manufacturing, and six sigma. Lean, Six Sigma, and the Theory of Constraints, the second strategy, perform better than the first way. The second integrated strategy improves both employee and employer knowledge of the performance metric. The second strategy boosts shareholder and employer confidence more effectively. Implementing this strategy thereby boosts the financial gains for the shareholder and the employee.

**Keywords:** Lean Management, Six Sigma, TOC, Ranking, Weightage, Standard Deviation.

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Technical Paper | Published: 04 October 2022

# Influence of CNT volume fractions on static and dynamic behavior of aluminum honeycomb-cored carbon-fiber-reinforced honeycomb sandwich structure

Shubham Chandra, Amit Amulani, Shweta B. Thomas,

**Sourabh Kumar Soni** & Benedict Thomas *Journal of the Brazilian Society of Mechanical Sciences and Engineering* 44, Article number: 507 (2022)340 Accesses | 3 Citations | [Metrics](#)

## Abstract

Sandwich structures made up of honeycomb composites have high-strength structural applications because of their low weight and high stiffness. High flexural stiffness is a major aspect in developing structures with several multilayer materials. Statistical tests analyze the behavior of loads to which a body can sustain without failure and dynamic behavior defines the overall failure modes and the vibration characteristics. In this study, static and dynamic responses of honeycomb sandwich

<https://link.springer.com/article/10.1007/s40430-022-03825-z>

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High Temperature Alloys: Manufacturing, Processing, and Repair

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# Phase Evolution of Novel MoNbSiTiW Refractory High-Entropy Alloy Prepared by Mechanical Alloying

[Om Prakash](#), [Rituraj Chandrakar](#), [Saurabh Chandraker](#), [K. Raja Rao](#), [Rajesh Kumar](#), [Anil Kumar](#)  & [Vikas Dubey](#)

*JOM* **74**, 3329–3333 (2022)474 Accesses | 3 Citations | [Metrics](#)

## Abstract

Refractory high-entropy alloys (RHEAs) are new types of material that have been developed for high-temperature applications. RHEAs should have enhanced high-temperature strength while maintaining a sufficient level of room-temperature toughness. The phase evolution of novel MoNbSiTiW RHEAs was investigated after mechanical alloying (MA) for 35 h. X-ray diffraction (XRD) was used to analyze the phase evolution, and analysis of particle morphologies was done using a scanning electron microscope (SEM) equipped with energy dispersive

# Natural Bamboo Fibre Composites, their properties and its Applications - A Review.

- **Source:** Journal of Production & Industrial Engineering (JPiE) . Jan-Jun2022, Vol. 3 Issue 1, p1-6. 6p.
- **Author(s):** Kumar, Roopesh; Rao, B. Laxshaman; **Ganguly, Abhijeet**; Purohit, Rajesh
- **Abstract:** In the present situation, there has been a fast consideration in innovative work in the Bamboo fibre composite field because of its better formability, copious, sustainable, financially etc. Bamboo fibre is a natural fibre that is easily degraded by microbes and potentially becomes an alternative fibre in the future due to its availability, which is abundantly cheap, and grows throughout the year and also isn't affected by the season The application of bamboo fibre is a solution for environmental issues and a business prospect of synthetic fibres that are expected to decrease in line with petroleum. With this work, various values of bamboo have been considered as the basis, mechanical and chemical properties have been given the with major focus. On the basis of these values and qualities, the characteristic and utility of bamboo have been told. The symptoms of use of bamboo have been considered.
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# A Simulation Of A Statcom-Control For Grid Connected Wind Energy System For Improving Reliability Of Power

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## ABSTRACT

Injection of the wind power into an electric grid affects the strength exceptional. The overall performance of the wind turbine and thereby electricity nice are determined on the basis of measurements and the norms observed according to the rule specified in International Electro-technical Commission trendy, IEC-61400. The have an impact on of the wind turbine within the grid gadget concerning the electricity pleasant measurements are-the active strength, reactive power, variation of voltage, flicker, harmonics, and electrical conduct of switching operation and these are measured consistent with national/global suggestions. The electricity exceptional problem mainly due to set up of wind turbine with the grid. In this proposed scheme STATIC COMPENSATOR (STATCOM) is attached at a factor of common coupling with a battery electricity storage machine (BESS) to mitigate the energy first-class problems. The battery strength garage is included to sustain the actual strength source below fluctuating wind power. The STATCOM manipulate scheme for the grid linked wind electricity era machine for power excellent development is simulated the usage of MATLAB/SIMULINK in power system block set. The effectiveness of the proposed scheme relieves the primary deliver source from the reactive power demand of the burden and the induction generator. The improvement of the grid co-ordination rule and the scheme for improvement in energy excellent norms as according to IEC-preferred at the grid has been supplied.

**Keywords:** STATCOM, BESS, reactive power, IEC-61400, grid.

## I. INTRODUCTION

Wind power is the conversion of wind energy into useful form, such as electricity, using wind turbines. Wind energy is directly used to crush grain or water pump. Wind energy is plentiful, renewable, widely distributed, cleans, and reduces greenhouse gas emissions when it displaces fossil-fuel-derived electricity. Today, modern energy industry faces a growing awareness regarding the impact of conventional power generation on the environment. Issues such as limited fossil fuel reserves, climate change due to CO<sub>2</sub> emissions, bring to attention alternative technologies to generate electricity in a more sustainable manner. The intermittency of wind seldom creates insurmountable problems when using wind power to supply a low proportion of total demand, but it presents extra costs when wind is to be used for a large fraction of demand. The latest technological advancements in wind energy conversion and an increased support from governmental and private institutions have led to increased wind power generation in recent years. Wind power is the fastest growing renewable source of electrical energy [1]. The wind power has increased in the past few years, hence it has become necessary to address problems associated with maintaining a stable electric power system that contains different sources of energy including hydro, thermal, coal, nuclear, wind, and solar. In the past, the total installed wind power capacity was a small fraction of the power system and continuous connection of the wind farm to the grid was not a major concern. The wind farm capacity is being continuously increased through the installation of more and larger wind turbines. Voltage stability and an efficient fault ride through capability are the basic requirements for higher penetration. Wind turbines have to be able to continue uninterrupted operation under transient voltage conditions to be in accordance with the grid codes [1]. Grid codes are certain standards set by regulating agencies. Wind power systems should meet these requirements for interconnection to the grid. Different grid code standards are established by different regulating bodies, but Nordic grid codes are becoming increasingly important. One of the major issues concerning a wind farm interconnection to a power grid concerns its impact on the power system [3]. Voltage instability problems occur in a power system that is not able to meet the reactive power demand.

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## Fake Review Prediction of Online Product: A Survey

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**Abstract :** Online reviews play a vital role in today's business and commerce. In the world of e-commerce, reviews are the best signs of success and failure. Businesses that have good reviews get a lot of free exposure on websites and pages that have good reviews show up at the top of the search results. Fake reviews are everywhere online. Online fake reviews are the reviews which are written by someone who has not actually used the product or the services. This paper presents fake review prediction of online products. The artificial intelligence based techniques is capable to extract the features and identified the trueness of the online fake reviews. The accuracy, error rate, precision is the key parameters and Python software can be used for implementation.

**IndexTerms** – Fake Review, AI, Prediction, Accuracy, Error, Data Mining.

### I. INTRODUCTION

With the widespread of fake online reviews, the detection of fake reviews has become a hot research issue. Despite the efforts of existing studies on fake review detection, the issues of imbalanced data and feature pruning still lack sufficient attention. To address these gaps, the present study proposes an ensemble model for the detection of fake online reviews. The model consists of four steps, and the first three steps are proposed to optimize the base classifiers: (i) Data resampling: We propose a novel way to address the data imbalance problem by combining the resampling and the grid search technique. (ii) Feature pruning: We propose an ablation study to drop unimportant features. (iii) Parameters optimization: We apply the grid search algorithm to determine suitable values of the relevant parameters for each base classifier. (iv) Classifier ensembling: We apply majority voting and stacking strategies to integrate the optimized base classifiers. The proposed data resampling method is also applied for the meta-classifier in the stacking ensemble model [1].

In e-commerce, user reviews can play a significant role in determining the revenue of an organisation. Online users rely on reviews before making decisions about any product and service. As such, the credibility of online reviews is crucial for businesses and can directly affect companies' reputation and profitability. That is why some businesses are paying spammers to post fake reviews. These fake reviews exploit consumer purchasing decisions. Consequently, the techniques for detecting fake reviews have extensively been explored in the past twelve years. However, there still lacks a survey that can analyse and summarise the existing approaches. To bridge up the issue, this survey paper details the task of fake review detection, summing up the existing datasets and their collection methods. It analyses the existing feature extraction techniques. It also summarises and analyses the existing techniques critically to identify gaps based on two groups: traditional statistical machine learning and deep learning methods [2].

The impact of online reviews on businesses has grown significantly during last year's, being crucial to determine business success in a wide array of sectors, ranging from restaurants, hotels to e-commerce. Unfortunately, some users use unethical means to improve their online reputation by writing fake reviews of their businesses or competitors. Previous research has addressed fake review detection in a number of domains, such as product or business reviews in restaurants and hotels. However, in spite of its economical interest, the domain of consumer electronics businesses has not yet been thoroughly studied [6]. Online reviews are one of the major factors for the customers to purchase any product or to get service from many sources of information that can be used to determine the public opinion on the products. Fake reviews will be published intentionally to drive the web traffic towards the particular products. These fake reviewers mislead the customers to distract the purchasers mind. Reviewers behaviors are extracted based the semantical analysis of his review content for the purpose of identifying the review as fake or not [7].

With evolution of 4G and availability of wireless Internet access at a much higher speed, makes it possible for developing countries to use E-commerce for getting product and services. As each and every company now-a-days have presence in online mode of marketing, getting the right product as well as service is also tough. This leads to the importance of online reviews on the Internet. For taking purchase as well as financial decisions, every individual have to depend on online reviews [9]. Although the